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possibilities given in the explanation for (1) could occur, since 460 miles in 4 hours gives an average speed of $\frac{460}{4} = 115$ miles per hour; NOT sufficient.

Assuming (1) and (2), each of the possibilities given in the explanation for (1) could occur. Therefore, (1) and (2) together are not sufficient.

The correct answer is E; both statements together are still not sufficient.

•

Tom, Jane, and Sue each purchased a new house. The average (arithmetic mean) price of the three houses was $120,000. What was the median price of the three houses?

1. The price of Tom's house was $110,000.
2. The price of Jane's house was $120,000.

Arithmetic Statistics

Let T, J, and S be the purchase prices for Tom's, Jane's, and Sue's new houses, respectively. Given that the average purchase price is 120,000, or $T + J + S = 3(120{,}000)$, determine the median purchase price.

1. Given $T = 110{,}000$, the median could be 120,000 (if $J = 120{,}000$ and $S = 130{,}000$) or 125,000 (if $J = 125{,}000$ and $S = 125{,}000$); NOT sufficient.
2. Given $J = 120{,}000$, the following two cases include every possibility consistent with $T + J + S = (3)(120{,}000)$, or $T + S = (2)(120{,}000)$.
   i. $T = S = 120{,}000$
   ii. One of T or S is less than 120,000 and the other is greater than 120,000.

   In each case, the median is clearly 120,000; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

A certain bookcase has 2 shelves of books. On the upper shelf, the book with the greatest number of pages has 400 pages. On the lower shelf, the book with the least number of pages has 475 pages. What is the median number of pages for all of the books on the 2 shelves?

1. There are 25 books on the upper shelf.
2. There are 24 books on the lower shelf.

Arithmetic Statistics

1. The information given says nothing about the number of books on the lower shelf. If there are fewer than 25 books on the lower shelf, then the median number of pages will be the number of pages in one of the books on the upper shelf or the average number of pages in two books on the upper shelf. Hence, the median will be at most 400. If there are more than 25 books on the lower shelf, then the median number of pages will be the number of pages in one of the books on the lower shelf or the average number of pages in two books on the lower shelf. Hence, the median will be at least 475; NOT sufficient.
2. An analysis very similar to that used in (1) shows the information given is not sufficient to determine the median; NOT sufficient.

Given both (1) and (2), it follows that there is a total of 49 books. Therefore, the median will be the 25th book when the books are ordered by number of pages. Since the 25th book in this ordering is the book on the upper shelf with the greatest number of pages, the median is 400. Therefore, (1) and (2) together are sufficient.

The correct answer is C; both statements together are sufficient.

•

In planning for a trip, Joan estimated both the distance of the trip, in miles, and her average speed, in miles per hour. She accurately divided her estimated distance by her estimated average speed to obtain an estimate for the time, in hours, that the trip would take. Was her estimate within 0.5 hours of the actual time that the trip took?

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hours, that the trip would take. Was her estimate within 0.5 hours of the actual time that the trip took?

1. Joan's estimate for the distance was within 5 miles of the actual distance.
2. Joan's estimate for her average speed was within 10 miles per hour of her actual average speed.

Arithmetic Applied Problems; Estimating

1. Given that the estimate for the distance was within 5 miles of the actual distance, it is not possible to determine whether the estimate of the time (that is, the estimated distance divided by the estimated average speed) was within 0.5 hour of the actual time because nothing is known about the estimated average speed; NOT sufficient.
2. Given that the estimate for the average speed was within 10 miles per hour of the actual average speed, it is not possible to determine whether the estimate of the time (that is, the estimated distance divided by the estimated average speed) was within 0.5 hours of the actual time because nothing is known about the estimated distance; NOT sufficient.

Taking (1) and (2) together, the table below shows two possibilities for the estimated distance, estimated average speed, actual distance, and actual average speed that are consistent with the information given in (1) and (2).

estimated distance	estimated speed	actual distance	actual speed
20 miles	$20\frac{\text{miles}}{\text{hour}}$	20 miles	$20\frac{\text{miles}}{\text{hour}}$
20 miles	$20\frac{\text{miles}}{\text{hour}}$	20 miles	$10\frac{\text{miles}}{\text{hour}}$

In the first row of the table, the estimate of the time and the actual time are each equal to 1 hour, and so the estimate of the time is within 0.5 hours of the actual time. In the second row of the table, the estimate of the time is 1 hour and the actual time is $\frac{20}{10} = 2$ hours, and so the estimate of the time is not within 0.5 hours of the actual time.

The correct answer is E; both statements together are still not sufficient.

Shipment	S1	S2	S3	S4	S5	S6
Fraction of the Total Value of the Six Shipments	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{3}{20}$	$\frac{2}{15}$	$\frac{1}{10}$

•

Six shipments of machine parts were shipped from a factory on two trucks, with each shipment entirely on one of the trucks. Each shipment was labeled either S1, S2, S3, S4, S5, or S6. The table shows the value of each shipment as a fraction of the total value of the six shipments. If the shipments on the first truck had a value greater than $\frac{1}{2}$ of the total value of the six shipments, was S3 shipped on the first truck?

1. S2 and S4 were shipped on the first truck.
2. S1 and S6 were shipped on the second truck.

Arithmetic Operations on Rational Numbers

Given that the shipments on the first truck had a value greater than $\frac{1}{2}$ of the total value of the 6 shipments, determine if S3 was shipped on the first truck.

To avoid dealing with fractions, it will be convenient to create scaled values of the shipments by multiplying each fractional value by 60, which is the least common denominator of the fractions. Thus, the scaled values associated with S1, S2, S3, S4, S5, and S6 are 15, 12, 10, 9, 8, and 6, respectively. The given information is that the scaled value of the shipments on the first truck is greater than $\left(\frac{1}{2}\right)(60) = 30$.

1. Given that the first truck includes shipments with scaled values 12 and 9, it may or may not be the case that S3 (the shipment with scaled value 10) is on the first truck. For example, the first truck could contain only S2, S3, and S4, for a total scaled value 12 + 10 + 9 = 31 > 30. Or the first truck could contain only S1, S2, and S4, for a total scaled value 15 + 12 + 9 = 36 > 30; NOT sufficient.

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2. Given that the second truck includes shipments with scaled values 15 and 6, the second truck cannot contain S3. Otherwise, the second truck would contain shipments with scaled values 15, 6, and 10, for a total scaled value 15 + 6 + 10 = 31, leaving at most a total scaled value 29 (which is not greater than 30) for the first truck; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

	Favorable	Unfavorable	Not Sure
Candidate M	40	20	40
Candidate N	30	35	35

•

The table above shows the results of a survey of 100 voters who each responded "Favorable" or "Unfavorable" or "Not Sure" when asked about their impressions of Candidate M and of Candidate N. What was the number of voters who responded "Favorable" for both candidates?

1. The number of voters who did not respond "Favorable" for either candidate was 40.
2. The number of voters who responded "Unfavorable" for both candidates was 10.

Arithmetic Sets

Let x, y, and z, respectively, be the numbers of voters who responded "Favorable," "Unfavorable," and "Not Sure" to both candidates. The table below gives a more complete tabulation of the survey results. For example, since 20 voters responded "Unfavorable" for Candidate M and y of those 20 voters also responded "Unfavorable" for Candidate N, it follows that $20-y$ of those 20 voters responded "Unfavorable" for Candidate M only. Determine the value of x, the number of voters who responded "Favorable" for both candidates.

	Favorable	Unfavorable	Not Sure
M and N	x	y	z
M only	$40-x$	$20-y$	$40-z$
N only	$30-x$	$35-y$	$35-z$

1. It is given that the number of voters who did not respond "Favorable" for either candidate was 40. Also, from the table above, the number of voters who responded "Favorable" for at least one candidate was $(40-x)+(30-x)+x=70-x$. Since there were 100 voters, it follows that $(70-x)+40=100$, which can be solved for a unique value of x; SUFFICIENT.
2. It is given that the number of voters who responded "Unfavorable" for both candidates was 10, or $y=10$. However, there is not enough information to determine the value of x. This can be seen by choosing different integer values for x between 0 and 30, inclusive, and using each value to complete the table above (choose $z=5$ for completeness, for example); NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

A school administrator will assign each student in a group of n students to one of m classrooms. If $3<m<13<n$, is it possible to assign each of the n students to one of the m classrooms so that each classroom has the same number of students assigned to it?

1. It is possible to assign each of $3n$ students to one of m classrooms so that each classroom has the same number of students assigned to it.
2. It is possible to assign each of $13n$ students to one of m classrooms so that each classroom has the same number of students assigned to it.

Arithmetic Properties of Numbers

Determine if n is divisible by m.

1. Given that $3n$ is divisible by m, then n is divisible by m if $m=9$ and $n=27$ (note that $3<m<13<n$, $3n=81$, and $m=9$, so $3n$ is divisible by m), and n is not divisible by m if $m=9$ and $n=30$ (note that $3<m<13<n$, $3n=90$, and $m=9$, so $3n$ is divisible by m); NOT sufficient.

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$3 < m < 13 < n$, $3n = 90$, and $m = 9$, so $3n$ is divisible by m); NOT sufficient.

2. Given that $13n$ is divisible by m, then $13n = qm$, or $\frac{n}{m} = \frac{q}{13}$, for some integer q.

Since 13 is a prime number that divides qm (because $13n = qm$) and 13 does not divide m (because $m < 13$), it follows that 13 divides q.

Therefore, $\frac{q}{13}$ is an integer, and since $\frac{n}{m} = \frac{q}{13}$, then $\frac{n}{m}$ is an integer. Thus, n is divisible by m; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

At a certain clothing store, customers who buy 2 shirts pay the regular price for the first shirt and a discounted price for the second shirt. The store makes the same profit from the sale of 2 shirts that it makes from the sale of 1 shirt at the regular price. For a customer who buys 2 shirts, what is the discounted price of the second shirt?

1. The regular price of each of the 2 shirts the customer buys at the clothing store is $16.
2. The cost to the clothing store of each of the 2 shirts the customer buys is $12.

Arithmetic Applied Problems

Solving this problem relies on the definition of profit: profit = selling price – cost.

1. Given that the regular price of each of the 2 shirts is $16, the discounted price cannot be uniquely determined because no information is given about the rate of discount, the amount of discount, or the cost; NOT sufficient.
2. Recall that profit = selling price – cost. Since the store makes the same profit from the sale of 2 shirts as it makes from the sale of 1 shirt at the regular price, the profit from the second shirt is $0. It follows that the selling price (that is, the discounted price) of the second shirt equals its cost.

Given that the cost of each of the shirts is $12, it follows that the discounted price of the second shirt is $12; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

What is the median number of employees assigned per project for the projects at Company Z?

1. 25 percent of the projects at Company Z have 4 or more employees assigned to each project.
2. 35 percent of the projects at Company Z have 2 or fewer employees assigned to each project.

Arithmetic Statistics

1. Although 25 percent of the projects have 4 or more employees, there is essentially no information about the middle values of the numbers of employees per project. For example, if there were a total of 100 projects, then the median could be 2 (75 projects that have exactly 2 employees each and 25 projects that have exactly 4 employees each) or the median could be 3 (75 projects that have exactly 3 employees each and 25 projects that have exactly 4 employees each); NOT sufficient.
2. Although 35 percent of the projects have 2 or fewer employees, there is essentially no information about the middle values of the numbers of employees per project. For example, if there were a total of 100 projects, then the median could be 3 (35 projects that have exactly 2 employees each and 65 projects that have exactly 3 employees each) or the median could be 4 (35 projects that have exactly 2 employees each and 65 projects that have exactly 4 employees each); NOT sufficient.

Given both (1) and (2), 100 – (25 + 35) percent = 40 percent of the projects have exactly 3 employees. Therefore, when the numbers of employees per project are listed from least to greatest, 35 percent of the numbers are 2 or less and (35 + 40) percent = 75 percent are 3 or less, and hence the median is 3.

The correct answer is C; both statements together are sufficient.

•

Last year, a certain company began manufacturing Product X and sold every unit of Product X that it produced. Last year, the company's total expenses for manufacturing Product X were equal to $100,000 plus 5 percent of the company's total revenue from all units of Product X sold. If the company made a profit on Product X last year, did the company sell more than 21,000 units of Product X last year?

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company sell more than 21,000 units of Product X last year?

1. The company's total revenue from the sale of Product X last year was greater than $110,000.
2. For each unit of Product X sold last year, the company's revenue was $5.

Algebra Applied Problems

Recall that Profit = Revenue – Expenses, and for this company, the profit on Product X last year was positive. Then, letting R represent total revenue, in dollars, from all units of Product X sold,
$\text{profit} = R - (100{,}000 + 0.05R) = 0.95R - 100{,}000$. So, $0.95R - 100{,}000 > 0$.

1. This says R > 110,000 but gives no information on how many units were sold or the price at which they were sold. Therefore, whether the company sold more than 21,000 units cannot be determined; NOT sufficient.

number sold	unit price	R, in dollars	number sold > 21,000?
1,200	$100	120,000	no
22,000	$10	220,000	yes

Note that a calculator is not needed to verify that the profit (which is given in dollars by 95% of $R - 100{,}000$) is positive for each of these two values of R. For example,

10% of 120,000	=	12,000	move decimal point 1 place to the left
5% of 120,000	=	6,000	5% is $\frac{1}{2}$ of 10%
120,000 – 6,000	=	114,000	$R - 5\%$ of $R = 95\%$ of R
114,000 – 100,000	=	14,000	95% of $R - 100{,}000$

2. This says the unit price was $5. If x represents the number of units sold, then $R = 5x$ and, by substitution, the requirement that the profit is positive becomes $0.95(5x) - 100{,}000 > 0$.

Therefore, the company sold more than 21,000 units; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

Beginning in January of last year, Carl made deposits of $120 into his account on the 15th of each month for several consecutive months and then made withdrawals of $50 from the account on the 15th of each of the remaining months of last year. There were no other transactions in the account last year. If the closing balance of Carl's account for May of last year was $2,600, what was the range of the monthly closing balances of Carl's account last year?

1. Last year, the closing balance of Carl's account for April was less than $2,625.
2. Last year, the closing balance of Carl's account for June was less than $2,675.

Arithmetic Statistics

It is given that the closing balance for May was $2,600 and the closing balance for each month either increases by $120 or decreases by $50 relative to the closing balance for the previous month, depending on when the withdrawals began. Therefore, the closing balance for June was either $2,720 or $2,550, and the closing balance for April was either $2,480 or $2,650. The table below shows these possibilities for 5 of the months that the withdrawals could have begun. The values in the last two rows of the table remain the same for later months that the withdrawals could have begun.

$$
\begin{aligned}
0.95(5x) - 100{,}000 &> 0 && \text{given} \\
0.95(5x) &> 100{,}000 && \text{add 100,000 to both sides} \\
0.95x &> 20{,}000 && \text{divide both sides by 5} \\
x &> \frac{20{,}000}{0.95} && \text{divide both sides by 0.95} \\
x &> \frac{20{,}000}{19} && 0.95 = \frac{95}{100} = \frac{19}{20}
\end{aligned}
$$

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$$x > \frac{19}{20} \qquad 0.95 = \frac{95}{100} = \frac{19}{20}$$

$$x > (20{,}000)\left(\frac{20}{19}\right) \qquad \text{invert divisor and multiply}$$

$$x > 20{,}000 + (20{,}000)\left(\frac{1}{19}\right) \qquad \frac{20}{19} = 1 + \frac{1}{19}$$

$$x > 20{,}000 + 1{,}000 \qquad \frac{1}{19} > \frac{1}{20}, (20{,}000)\left(\frac{1}{20}\right) = 1{,}000$$

$$x > 21{,}000 \qquad \text{add}$$

Withdrawals began	Close April	Close May	Close June
Apr 15	$2,650	$2,600	$2,550
May 15	$2,650	$2,600	$2,550
Jun 15	$2,480	$2,600	$2,550
Jul 15	$2,480	$2,600	$2,720
Aug 15	$2,480	$2,600	$2,720

1. Given that the closing balance for April was less than $2,625, it follows from the table above that the withdrawals could have begun in June or in any later month. For each of these possibilities, the balance at the beginning of January was $2,600 – 5($120) = $2,000, since each of the 5 transactions made before the end of May were $120 deposits. Therefore, the range of the monthly closing balances of the account can vary because the range will depend on how many of the transactions made after May were $120 deposits and how many were $50 withdrawals; NOT sufficient.
2. Given that the closing balance for June was less than $2,675, it follows from the table above that the withdrawals could have begun in June or in any earlier month. For each of these possibilities, the balance at the end of December was $2,600 – 7($50) = $2,250, since each of the 7 transactions made after the end of May were $50 withdrawals. Therefore, the range of the monthly closing balances of the account can vary because the range will depend on how many of the transactions made before May were $120 deposits and how many were $50 withdrawals; NOT sufficient.

Taking (1) and (2) together, withdrawals started in June or later by (1) and withdrawals started in June or earlier by (2), so withdrawals started in June. Therefore, each of the monthly closing balances of the account is known, and thus the range of these monthly closing balances can be determined.

The correct answer is C; both statements together are sufficient.

•

What amount did Jean earn from the commission on her sales in the first half of 1988?

1. In 1988, Jean's commission was 5 percent of the total amount of her sales.
2. The amount of Jean's sales in the second half of 1988 averaged $10,000 per month more than in the first half.

Arithmetic Applied Problems

Let A be the amount of Jean's sales in the first half of 1988. Determine the value of A.

1. If the amount of Jean's sales in the first half of 1988 was $10,000, then her commission in the first half of 1988 would have been (5%)($10,000) = $500. On the other hand, if the amount of Jean's sales in the first half of 1988 was $100,000, then her commission in the first half of 1988 would have been (5%)($100,000) = $5,000; NOT sufficient.
2. No information is given that relates the amount of Jean's sales to the amount of Jean's commission; NOT sufficient.

Given (1) and (2), from (1) the amount of Jean's commission in the first half of 1988 is (5%)A. From (2) the amount of Jean's sales in the second half of 1988 is $A + \$60{,}000$. Both statements together do not give information to determine the value of A.

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The correct answer is E; both statements together are still not sufficient.

•

In a certain business, production index p is directly proportional to efficiency index e, which is in turn directly proportional to investment index i. What is p if $i = 70$?

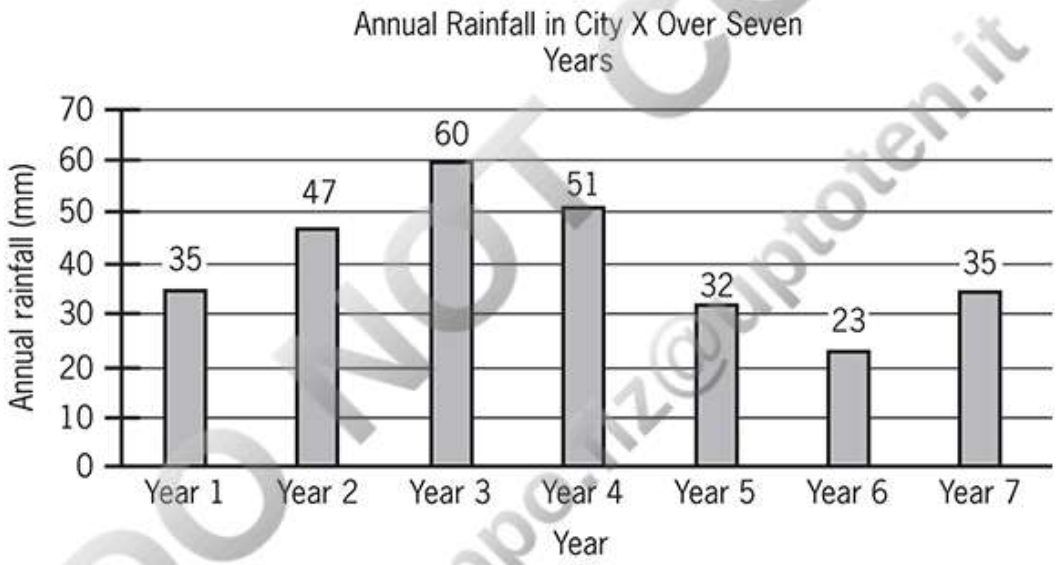
1. $e = 0.5$ whenever $i = 60$.
2. $p = 2.0$ whenever $i = 50$.

Arithmetic Proportions

1. This gives only values for e and i, and, while p is directly proportional to e, the nature of this proportion is unknown. Therefore, p cannot be determined; NOT sufficient.
2. Since p is directly proportional to e, which is directly proportional to i, then p is directly proportional to i. Therefore, the following proportion can be set up: $\frac{p}{i} = \frac{2.0}{50}$. If $i = 70$, then $\frac{p}{70} = \frac{2.0}{50}$. Through cross multiplying, this equation yields $50p = 140$, or $p = 2.8$; SUFFICIENT.

The preceding approach is one method that can be used. Another approach is as follows: It is given that $p = Ke = K(Li) = (KL)i$, where K and L are the proportionality constants, and the value of $70KL$ is to be determined. Statement (1) allows us to determine the value of L but gives nothing about K, and thus (1) is not sufficient. Statement (2) allows us to determine the value of KL, and thus (2) is sufficient.

The correct answer is B; statement 2 alone is sufficient.



•

The annual rainfall in a city over the past 50 years is its historical average rainfall (HAR). The graph above shows the annual rainfall in City X over a seven-year period. All figures are rounded to the nearest integer.

Is it true that the rounded historical average rainfall (HAR) for City X over the past 50 years is 47 mm or greater?

1. In the seven-year period in City X, the annual rainfall was greater than the HAR in exactly two of the consecutive years.
2. In the seven-year period in City X, the annual rainfall was less than the HAR in exactly four of the years.

Inference

1. According to this statement, in the seven-year period in City X depicted in the graph—a period that falls sometime during the past 50 years—the average annual rainfall was greater than the HAR in only two consecutive years. Those two years must be years 3 and 4, because they are the two years with the greatest annual rainfall. That means that none of the remaining years' average annual rainfalls was greater than the rounded HAR. It can be inferred, then, that the rounded HAR for City X over the past 50 years was at least as great as the average annual rainfall of year 2, that is, 47 mm; SUFFICIENT.

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2. According to this statement, in the seven-year period in City X depicted in the graph—a period that falls sometime during the past 50 years—the average annual rainfall was less than the HAR in exactly four of the years. It can be inferred that those years are years 1, 5, 6, and 7, so the rounded HAR must be at least 36 but no more than 47. Thus, we can infer that the HAR may, but need not be, as great as 47 mm; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

Five coffee-chicory blends—J, K, L, M, and N—are all priced differently. The blends are such that the greater the proportion of coffee, the more expensive the blend, and the greater the proportion of chicory, the stronger the flavor of the blend. If the five blends were ranked in order of cost per kilogram, which would fall in the middle?

1. Blend L has a higher proportion of coffee than the other blends except for Blend M.
2. Blend K and Blend N both have a stronger flavor than Blend J.

Inference

From the stimulus we know that there are five different coffee-chicory blends, J, K, L, M, and N, and they are differentiated by their relative proportions of coffee to chicory. The greater the proportion of coffee, the greater the price. The greater the proportion of chicory, the stronger the flavor and the less the proportion of coffee. We are told that no two blends have the same prices. We can infer that no two blends are equally strong in flavor. Given those two inferences, we can also infer that the stronger the flavor, the less expensive the blend.

1. We can infer that Blend M is the most expensive blend and Blend L is the next most expensive. We can infer that Blends J, K, and N are the three least expensive (and thus the three strongest in flavor), but we do not know in what order, so we don't know which would fall in the middle if the five blends were ranked in order of cost per kilogram; NOT sufficient.
2. We can infer that Blend K and Blend N have stronger flavors than Blend J, and thus are less expensive than Blend J. Therefore, Blend J could be either the most, second-most, or third-most expensive, and thus Blend J might fall in the middle if the five blends were ranked in order of cost per kilogram, but the information in the stimulus and (2) is not sufficient to determine whether it does fall in the middle; it could also be the most or second-most expensive; NOT sufficient.

From the stimulus and (1), we can infer that Blend J is either the third-most, fourth-most, or fifth-most expensive blend. Also, from the stimulus and (2), we can infer that Blend J is either the most, second-most, or third-most expensive blend. Therefore, from the stimulus and (1) and (2) taken together, we can infer that Blend J is the third-most expensive blend, which means that it would fall in the middle if the five blends were ranked in order of cost per kilogram.

The correct answer is C; both statements together are sufficient.

•

Last summer, as with every summer, Oshin and Reena made exactly one long trip together to a town neither of them had visited before. For their long summer trips, Oshin and Reena have always chosen to visit a town that Reena would have liked to visit—unless Oshin had been to that town before. Reena had visited Town F and Town G before, and Oshin had never visited Town J or Town K. Did Oshin and Reena visit one of these four towns last summer?

1. Reena had never visited Town J, and Oshin had never visited Town F.
2. Oshin would have liked to visit either Town G or Town K.

Inference

We can infer that they did NOT visit Town F or Town G, because the town they visited had never been visited by either one of them, and Reena had visited Towns F and G. We are told in the stimulus that Oshin has never visited Town J or Town K, but we are not told whether Reena has visited them or has any interest in visiting them.

1. We already know that Oshin and Reena did not visit Town F (because the prior information had told us that Reena had visited there before). This statement neither rules out nor supports that Oshin and Reena visited Town J. We would need to know whether Reena was interested in visiting Town J, but we do not; NOT sufficient.
2. We have been told that Oshin and Reena have always visited a town that Reena would like to visit, so long as Oshin has not visited the town. (2) does not tell us whether Reena was interested in visiting Town G or Town K, only that Oshin would have liked to; NOT sufficient.

Neither (1) nor (2), either in isolation or together, gives us any information one way or the other about what town

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Neither (1) nor (2), either in isolation or together, gives us any information one way or the other about what town Reena would have liked to visit, but that is information that we would need to answer the question, so statements 1 and 2 taken together are not sufficient.

The correct answer is E; both statements together are still not sufficient.

•

For a new project, an organization is creating a three-member team from among its employees. Selection of members for the team conforms to exactly two rules: no selected member can be working on more than one other project at the time of selection, and at least one of the selected members must have an MBA degree. The organization has selected Paula and Quincy for the team. Neither of these will be replaced before the third member is selected. The organization is about to make the third selection. If the organization adheres to the selection rules for committee membership, is it possible that Rubin will be selected for the team?

1. Quincy has an MBA degree, and Rubin is currently working on only one other project.
2. Rubin is one of only two employees in the organization who have an MBA degree.

Inference

Paula and Quincy were selected to be members, and they remain, on the team. Given the two selection rules, it will be possible for Rubin to be selected so long as he is working on no more than one other project at the time he is selected and either he or at least one of the two already-selected members has an MBA.

1. This statement informs us that the two requirements are met, namely, that either Rubin or one of the already-selected members has an MBA (we are told Quincy has one) and that Rubin is currently working on only one other project; SUFFICIENT.
2. This statement informs us that one of the two requirements will be met if Rubin is selected—Rubin has an MBA—but we do not know from this statement alone whether Rubin meets the other requirement for his selection, namely, whether he is working on at most one other project; NOT sufficient.

The correct answer is A;statement 1 alone is sufficient.

•

All the T-shirts a clothing store stocks are plain, and they are all in one of exactly four colors: blue, green, red, and yellow. On any given day, the store sells about twice as many blue T-shirts as it does green T-shirts. Last week, did the store sell more blue T-shirts than red T-shirts?

1. Last week, the store sold fewer blue T-shirts than red T-shirts and green T-shirts combined.
2. Last week, the store sold more blue T-shirts than half the total number of red T-shirts and yellow T-shirts combined.

Inference

All the information we are given is that the store sells only blue, green, red, and yellow T-shirts and on any given day it sells about twice as many blue T-shirts as green T-shirts.

1. From (1) we can infer only that last week the store sold at least slightly more red T-shirts than green T-shirts, but this is compatible with the store having sold anywhere from only slightly more red T-shirts than half the number of blue T-shirts it sold to many more red T-shirts than blue T-shirts; NOT sufficient.
2. From (2) we can infer only that the store sold at most roughly twice as many red T-shirts as blue T-shirts (in which case the number of yellow T-shirts sold would be close to zero). But (2) is also compatible with the store having sold only as many red T-shirts as a tiny fraction of the number of blue T-shirts it sold, so long as the number of yellow T-shirts was sufficiently great; NOT sufficient.

From (1) and (2) combined, we can infer only that the number of red T-shirts sold last week was somewhere in the range of approximately one-half the number of blue T-shirts sold to approximately twice the number of blue T-shirts sold.

The correct answer is E; both statements together are still not sufficient.

Month	Forecast for Units of Product Z
January	12
February	16

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March	16
April	20
May	20
June	24

•

The table above shows the sales forecast from January to June of last year, by the sales team of Company X, for the number of units of Product Z each month.

Was the actual number of units of Product Z sold in March less than the number sold in April?

1. Twelve units of Product Z were sold in March.
2. From January through June of last year, there was a constant, linear relationship between the forecast and actual number of units of Product Z sold.

Inference

Note that the graph only tells us what the sales team had forecast for March and April (16 and 20 units sold, respectively), not the actual units sold.

1. This statement tells us only the actual number of units sold in March but nothing about the actual number of units sold in April. If the actual sales varied from the forecast sales in a random way from month to month, then it is possible that March's sales were less than, more than, or the same as April's sales; NOT sufficient.
2. Because this statement tells us that there is an unchanging linear relationship between the forecast and the actual number of units of Product Z sold through the months given in the graph, we know that for any month X given in the graph, if another month Y was forecast to have lower sales than X, then the actual sales of Y were lower than the sales of X. Therefore, since the forecast for units sold for March was lower than the forecast for units sold for April, then the actual sales for March were lower than the actual sales for April; SUFFICIENT.

The correct answer is B;statement 2 alone is sufficient.

•

Chris was in an electronic appliances store to buy an LED computer monitor. Chris's purchase criteria included screen size, which must be at least 32 inches, and price, which must not exceed $400. A friend made two recommendations: Monitor A and Monitor B. Chris liked both monitors. Did Chris more likely buy Monitor A, Monitor B, or neither?

1. Monitor A's screen size was more than 32 inches, and it cost less than Monitor B.
2. Monitor B cost $390, and its screen size was less than that of Monitor A.

Inference

Chris is in a store shopping for an LED computer monitor that must meet at least two criteria—the monitor must have a screen size of at least 32 inches and cost no more than $400. Note that we are not told whether these are Chris's only two criteria; there may be others. Chris likes both Monitor A and Monitor B, so presumably they both meet his criteria regarding screen size and price.

1. From this statement we can be certain that Monitor A's screen size is sufficiently large, but we cannot infer from it whether Monitor A meets Chris's price requirement. We cannot infer anything about Monitor B's screen size, and it may or may not be the case that its screen size is enough larger than Monitor A's that Chris would be willing to pay more for it, assuming neither monitor is more than $400; NOT sufficient.
2. From this statement we can conclude that Monitor B meets Chris's price criterion, but it does not allow us to infer anything about Monitor A's price—though it does inform us that Monitor B has a smaller screen size than Monitor A. It does not, however, give us information sufficient to infer whether either of the monitors meets his screen size requirement; NOT sufficient.

It might seem that from (1) and (2) taken together, we can infer that Chris would prefer Monitor A to Monitor B and would therefore choose to purchase Monitor A. After all, we can infer from those statements together that Monitor A meets both the screen size requirement and the price requirement (because we can infer that Monitor A costs less than $390) and that Monitor A is both larger and less expensive than Monitor B. But remember that there may be other criteria that Chris may use for assessing which monitor to purchase besides the two criteria mentioned, which might make Monitor B preferable to Monitor A. Furthermore, none of the given information confirms that Monitor A or Monitor B are LED monitors, the kind we are told Chris desires. Furthermore, there may be a third monitor that Chris